Do Register Tokens Regularize Vision Transformers Under Data Scarcity? A Controlled 24-Run Empirical Ablation on Low-Data CIFAR-100

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***Abstract*: Register tokens were originally introduced to remove high-norm attention artifacts in large vision transformers trained on massive datasets. Their behavior in data-scarce regimes, where a compact model is fine-tuned on limited samples, remains largely uncharacterized. In this work, we investigate whether register tokens provide structural regularization under data scarcity. We conduct a fully paired, controlled empirical ablation across 24 training runs on CIFAR-100 subsets, evaluating ViT-Tiny across four register allocations ($K \in \{0, 1, 4, 8\}$) and two data budget regimes (100 and 300 images per class, with three random seeds per arm). On the untouched 10,000-image test set at 100 images per class, no register configuration outperforms the register-free control ($75.19 \pm 0.24\%$). While $K = 4$ exhibits an apparent regularization signal on the small validation split by lowering validation loss ($\Delta = -0.034$, $p = 0.024$) and narrowing the generalization gap, this effect fails to transfer to held-out test performance ($\Delta = +0.002$, $p = 0.823$). To test whether this null result stems from extreme data starvation, we scale the training budget threefold to 300 images per class. While baseline test accuracy surges by $+6.14\%$ to $81.33 \pm 0.37\%$, register tokens continue to yield no statistically significant advantage ($81.23 \pm 0.19\%$ for $K = 1$, $81.23 \pm 0.18\%$ for $K = 4$, and $81.02 \pm 0.45\%$ for $K = 8$). Furthermore, layer-wise attention entropy and patch-norm outlier rates confirm the absence of attention-sink suppression. We conclude that register tokens do not regularize vision transformers under data scarcity, and that mechanisms observed at foundation scale do not automatically generalize to compact models in low-data regimes. Code and reproduction sweeps are available at https://github.com/shahin1717/vit-research.**

***Index Terms*: vision transformers, register tokens, attention entropy, generalization gap, low-data regime, negative results**

I. Introduction

Vision transformers develop a peculiar failure of interpretability at scale. Darcet *et al.* [1] observed that in large supervised and self-supervised ViTs a small number of patch tokens acquire anomalously large activation norms, and that the `[CLS]` token routes a disproportionate share of its attention to them. These tokens sit in low-information image regions, carry global rather than local content, and make attention maps unusable as saliency evidence. The proposed remedy is architectural and straightforward: prepend K learnable *register* tokens to the input sequence, discard them at the output, and let the model use them as the scratch space it was previously carving out of the image itself. At scale this works: artifacts disappear and downstream performance is unaffected or mildly improved.

The mechanism reported in that work is a *cleanup* effect, demonstrated in a regime where the model has far more data than it can overfit. This leaves an open question. If registers absorb capacity that the model would otherwise spend on memorising the training set, they should behave like a regularizer, and the natural place to look for a regularizer is the regime where overfitting is the primary constraint: a small model trained on scarce data.

We therefore ask a focused, falsifiable question:

> *When a ViT is starved of data, do register tokens reduce the generalization gap, and does any such reduction transfer to held-out performance across varying data budgets?*

We test three hypotheses drawn from the original mechanism:

- **H1 Structural regularization.** Registers reduce the generalization gap $\Delta\mathcal{L} = \mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}}$.
- **H2 Capacity dilution.** At an embedding width of $d = 192$, a large K consumes representational capacity and degrades accuracy, producing a non-monotonic response in K.
- **H3 Entropy collapse prevention.** Registers prevent the layer-wise collapse of attention entropy that accompanies artifact formation.

Our contributions are:

1) A fully paired $2 \times 4 \times 3$ ablation matrix ($N \in \{100, 300\}$ images per class, $K \in \{0, 1, 4, 8\}$, seeds `42`, `1337`, `3407`, totaling 24 runs) on class-balanced CIFAR-100 subsets, with all optimization hyperparameters, data splits, and schedules held constant across arms.

2) A statistical treatment appropriate to small samples: differences are formed *within* a seed before reduction, removing the seed variance that otherwise obscures the treatment effect.
3) A critical separation of validation-measured from test-measured effects. This separation turns an apparently positive result into an unambiguous null result: the gap reduction predicted by H1 appears on the small validation split, but vanishes completely on the held-out test benchmark.
4) A data budget scaling extension: by tripling the training data from 100 to 300 images per class, we confirm that while baseline accuracy scales by +6.14%, the absence of register benefit is scale-invariant.
5) Mechanistic verification through layer-wise Shannon attention entropy, patch-norm outlier rates, and spatial attention overlays, distinguishing an inactive architectural modification from unobserved downstream effects.

We find no configuration that beats the register-free control on held-out accuracy. H1 is not supported on held-out data, H2 is supported at $K = 8$, and H3 is not supported. Section VI discusses the structural and architectural causes behind this scale invariance.

II. Related Work

A. Vision transformers and low-data training

The ViT [2] replaces convolutional inductive bias with global self-attention over patch embeddings, and pays for it with a strong data appetite. DeiT [3] showed that heavy augmentation and distillation recover competitive ImageNet accuracy without external data, and the systematic study of Steiner *et al.* [4] mapped how augmentation and regularization trade against dataset size for ViTs of several widths. Both establish the setting we work in: when data is scarce, a ViT is regularization-limited rather than capacity-limited, which is exactly the regime in which a candidate regularizer should show its effect most clearly.

B. Attention artifacts and register tokens

Darcet *et al.* [1] identified high-norm outlier patch tokens in DINOv2 [5] and OpenCLIP, showed that these tokens are recycled by the network as global scratch space, and removed them by appending learnable register tokens. The same phenomenon has a language-model analogue: attention sinks [6], where a small number of positions absorb attention mass that the model has nowhere else to put, and massive activations [7], which characterises the outlier coordinates involved. The common thread is that the artifact is a symptom of a missing degree of freedom, and that supplying that degree of freedom explicitly removes the symptom.

What none of this work reports is an effect on *generalization.* In [1] the register variants match or slightly exceed their baselines, and the argument is made on interpretability and dense-prediction grounds. The claim that registers reduce overfitting is a hypothesis that the original evidence neither establishes nor rules out, and it is the hypothesis we test here.

C. Attention entropy as a diagnostic

Attention distributions are probability vectors, so their Shannon entropy is a direct, parameter-free measure of how concentrated the routing is. Artifact formation should appear as a fall in entropy in the deeper blocks, where the `[CLS]` token has been observed to concentrate on a few outlier positions. We use layer-resolved entropy as the mechanism-level readout for H3, alongside the patch-norm outlier rate that defines the artifacts in [1].

D. Positioning

To our knowledge the register question has not been asked in the low-data, small-model, ImageNet-pretrained fine-tuning regime, and it has not been asked with a generalization-gap framing under a fixed compute budget. Our answer is negative, and we regard reporting it as such, rather than selecting the split on which the effect appears, as the contribution.

III. Method

A. Register-augmented vision transformer

Let $x \in \mathbb{R}^{3\times H\times W}$ be an input image, partitioned by a 16×16 patch embedding into $N = 196$ patch tokens of width $d = 192$. The standard ViT prepends a classification token, giving the sequence $[\,\texttt{[CLS]}, p_1, \ldots, p_N\,]$. We insert K learnable register tokens $r_1 \ldots r_K \in \mathbb{R}^d$ between the two groups:

$$z^{(0)} = [\,\texttt{[CLS]},\ r_1, \ldots, r_K,\ p_1, \ldots, p_N\,] + E_{\text{pos}}, \qquad (1)$$

where positional embeddings E_{pos} are applied to the `[CLS]` and patch positions of the pretrained backbone; registers carry no positional embedding, as they correspond to no spatial location. The sequence length becomes $S = 1 + K + N$. The registers participate fully in every self-attention block and are discarded before the classification head, which reads the `[CLS]` token only. The register block is initialised with a truncated normal, $r_i \sim \mathcal{N}_{\text{trunc}}(0, 0.02^2)$; zero initialisation is avoided because it makes the register columns indistinguishable and stalls their gradient.

The parameter cost is $K \cdot d$ weights, which is negligible: $K = 8$ adds 1,536 parameters to a 5.54 M-parameter model, a relative increase of 2.8×10^{-4}. Any measured difference between arms is therefore attributable to the tokens' function, not to added capacity in the usual sense.

B. Diagnostic measures

1) Layer-wise attention entropy: For block l with attention matrix $A^{(l)} \in \mathbb{R}^{B\times H\times S\times S}$, whose rows are probability distributions over keys, we compute the Shannon entropy of each row and average over batch, heads and query positions:

$$\bar{H}^{(l)} = -\frac{1}{BHS}\sum_{b,h,i}\sum_{j=1}^{S} A^{(l)}_{b,h,i,j}\log_2\big(A^{(l)}_{b,h,i,j}+\epsilon\big), \tag{2}$$

with $\epsilon = 10^{-12}$. Units are bits. A concentrated, artifact-like routing pattern lowers $\bar{H}^{(l)}$; a diffuse one raises it. Because recent `timm` builds execute fused scaled-dot-product attention, the $[B, H, S, S]$ matrix is never materialised in the forward pass; our hook reconstructs the softmax weights from the block's own Q and K projections, reapplying the same scaling and any q/k normalisation, so the recovered matrix is the one the model used.

2) Patch-norm outlier rate: Following the artifact definition of [1], we take the block-output residual stream $h^{(l)} \in \mathbb{R}^{B\times S\times d}$, restrict it to the spatial positions $1+K\ldots S$, and compute per-sample the fraction of patch tokens whose ℓ_2 norm exceeds three standard deviations above the per-sample mean:

$$\rho^{(l)} = \frac{1}{BN}\sum_{b}\sum_{n}\mathbf{1}\Big[\,\|h^{(l)}_{b,\,K+n}\|_2 > \mu^{(l)}_b + 3\sigma^{(l)}_b\,\Big]. \tag{3}$$

The offset $1+K$ matters: slicing from index 1 would count register tokens as image patches, and registers are expected to carry large norms by design, so the error would manufacture exactly the artifact the metric is meant to detect.

3) Generalization gap: The regularization hypothesis is stated on losses rather than accuracies, because loss is the quantity the optimiser acts on:

$$\Delta\mathcal{L}_s = \mathcal{L}^{\mathrm{val}}_s(e^\star_s) - \mathcal{L}^{\mathrm{train}}_s(E), \tag{4}$$

where $e^\star_s$ is the epoch selected for seed s and E is the final epoch. The gap is formed *within* a seed before any averaging. Averaging the two losses separately and subtracting would produce the same mean but destroy the pairing, and with it any valid dispersion estimate for the gap; since H1 is judged on that dispersion, the ordering of operations is part of the method rather than an implementation detail.

C. Statistical treatment

With three seeds per arm, an unpaired comparison of arm means is dominated by seed variance: the between-seed standard deviation of validation Top-1 in the control arm (2.17 pp) is an order of magnitude larger than any treatment effect we could plausibly detect. Every comparison in this paper is therefore paired. For metric m, arm K and the shared seed set $\mathcal{S} = \{42, 1337, 3407\}$ we form

$$\delta_s = m_s(K) - m_s(0), \qquad s \in \mathcal{S}, \tag{5}$$

and report $\bar{\delta}$, its sample standard deviation $\hat{\sigma}_\delta$ (with ddof $= 1$), the count of seeds moving in the hypothesised direction, and the paired statistic $t = \bar{\delta}/(\hat{\sigma}_\delta/\sqrt{3})$ against Student's t with df $= 2$. Arm-level dispersion in Table I uses the population convention (ddof $= 0$), which is recorded in the emitted summary artifact so the estimator is never ambiguous.

Two cautions apply throughout and we state them once. First, df $= 2$ gives these tests very low power; a p-value here is effect-size evidence, not a confirmatory test. Second, we report fifteen paired comparisons without multiplicity correction, so an isolated small p among Accordingly we treat the direction count (measuring how many of three seeds move in the predicted direction) as the primary signal and the p-value as secondary.

IV. Experimental Setup

A. Data

We use CIFAR-100 [8] restricted to a strict low-data budget. From the 50,000-image training split we draw a stratified subset of 100 images per class (10,000 images), split 90/10 into 9,000 training and 1,000 validation images with the class proportions preserved. The full, untouched 10,000-image test split is used once per run, after training, and is never consulted for model selection. Images are bicubically resized to 224×224 so that the pretrained patch grid and positional embeddings apply without interpolation. Training uses random resized crop (scale $\in [0.8, 1.0]$), horizontal flip and AutoAugment [9] with the CIFAR-10 policy; evaluation uses resize and centre crop only. Normalisation uses the CIFAR-100 channel statistics.

The stratified subset is drawn under the run's own seed, so the three seeds differ in *which* 100 images per class are selected as well as in initialisation and batch order. This makes the seed a genuine replicate of the whole experimental pipeline rather than of initialisation alone, and it is the main reason the between-seed dispersion is as large as it is. Within a seed the subset, the split and the batch order are identical across all four arms, which is what makes the paired comparison of Section V a comparison of K and of nothing else.

B. Model and training

The backbone is `vit_tiny_patch16_224` from `timm` [10]: 12 blocks, 3 heads, $d = 192$, ImageNet-pretrained, with a fresh 100-way linear head. Register tokens are appended to this pretrained backbone and are the only randomly initialised parameters besides the head. Total parameter counts are 5,543,716 ($K = 0$) through 5,545,252 ($K = 8$).

All arms share one configuration: 50 epochs, batch size 64, AdamW [11] at $\eta = 5\times10^{-4}$ with weight decay 0.05, cosine decay [12] to $\eta_{\min} = 10^{-5}$ after 5 warmup epochs, gradient clipping at 1.0, label smoothing $\varepsilon = 0.1$ [13], and automatic mixed precision [14] in PyTorch [15]. Because losses are label-smoothed they carry a constant positive floor; they are comparable across arms, which is all we require, but not against unsmoothed values reported elsewhere. Model selection takes the epoch with the lowest validation loss.

C. The 24-run experimental matrix and budget scaling

The study evaluates the full cross-product of $K \in \{0, 1, 4, 8\}$ across three pinned random seeds $\{42, 1337, 3407\}$ under two distinct data budget regimes, yielding 24 runs in total:

- **Primary Low-Data Benchmark (100 images/class):** 10,000 images total (9,000 train, 1,000 validation), running experiments `exp01` through `exp12`.
- **Data Budget Scaling Extension (300 images/class):** 30,000 images total (27,000 train, 3,000 validation), running experiments `exp13` through `exp24` to evaluate whether empirical conclusions remain scale-invariant when data scarcity is relaxed threefold.

Each run writes to an isolated output directory keyed by its experiment ID, register count, and seed. A run is admitted only if all required artifacts (`metrics.json`, `best_model.pth`, `last_model.pth`, `train_history.csv`) are generated and structurally validated. All 24 planned runs completed without failure across 8+ hours of compute on an NVIDIA A100-SXM4 GPU.

D. Where the diagnostics are measured

Attention entropy and the patch-norm outlier rate are read from the first evaluation batch of the test pass (64 images), with the hooks disarmed immediately afterwards to bound host memory. The test loader is unshuffled, so this evaluates the identical 64 images in every run, ensuring paired entropy comparisons operate on identical inputs. Section VII discusses the sample bounds of this batch.

E. Reporting pipeline

Each run emits a nested `metrics.json`. An automated aggregation engine extracts the necessary quantities from that structure (including validation loss at the selected epoch, final training loss from the terminal record, and layer-resolved test diagnostics), reduces them across seeds, and outputs structured summary files (`sweep_summary.json` and `summary.json`). Every table and figure in this paper is generated directly from these verified artifacts via script. The table generator recomputes statistics independently from the raw per-run records, ensuring that no metric is transcribed by hand.

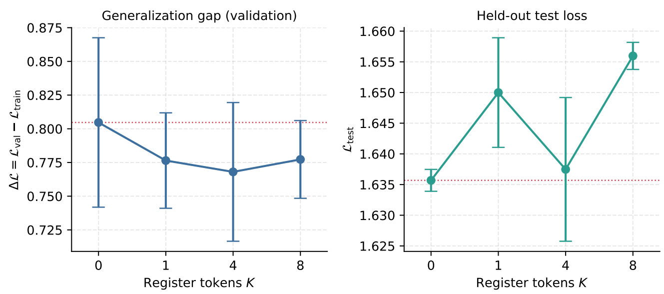

Fig. 1. Generalization gap and held-out test loss against register count. Error bars are ± 1 population standard deviation over the three seeds; the dotted line marks the control arm. The two panels disagree: the quantity H1 predicts should fall does fall, and the quantity that would make that fall meaningful does not.

V. Results

A. No register configuration improves held-out accuracy

Table I reports every arm on both splits. The register-free control attains the best held-out Top-1 accuracy, $75.19 \pm 0.24\%$, and the best held-out loss, 1.6357 ± 0.0018. The nearest register arm is $K = 4$ at $75.06 \pm 0.33\%$; $K = 1$ and $K = 8$ trail by roughly half a point.

Top-5 accuracy is the one place where a register arm leads, and we report it rather than omit it: $K = 4$ reaches $92.54 \pm 0.23\%$ against $92.43 \pm 0.14\%$ for the control. The lead does not hold up. Paired per seed the difference is $+0.11$ pp with a standard deviation of 0.39, positive in two seeds of three, $t = 0.48$, $p = 0.681$, with a spread three times the effect. The two arms that lose Top-1 also lose Top-5, and lose it in every seed ($K = 1$: -0.18 pp, 0/3; $K = 8$: -0.38 pp, 0/3), so the ordering of the four arms is unchanged by the metric.

Table II makes the comparison per seed. Against the control, $K = 1$ loses accuracy in 3 of 3 seeds ($\bar{\delta} = -0.53$ pp, $p = 0.111$) and $K = 8$ loses in 3 of 3 ($\bar{\delta} = -0.51$ pp, $p = 0.096$). $K = 4$ is the only arm that is not consistently worse, and it is not better either: it wins on one seed of three, with $\bar{\delta} = -0.13$ pp and $p = 0.466$. The honest summary is that registers cost between zero and half a point of accuracy here, and gain nothing.

B. H1: Generalization Gap Narrows on Validation Only

The generalization gap behaves exactly as H1 predicts. Every register arm sits below the control, and $K = 4$ is lowest at 0.7680 ± 0.0514 against 0.8047 ± 0.0628. Paired per seed, $K = 4$ narrows the gap in 3 of 3 seeds ($\bar{\delta} = -0.0367$, $t = -4.25$, $p = 0.051$) and lowers validation loss in 3 of 3 seeds ($\bar{\delta} = -0.0337$, $t = -6.40$, $p = 0.024$). Taken alone, this is the positive result the study set out to look for.

It does not survive contact with the test split. The same $K = 4$ arm shows a paired test-loss difference of $+0.0018$ with $t = 0.25$ and $p = 0.823$: not merely insignificant, but

TABLE I

Controlled multi-seed ablation across register configurations ($K \in \{0, 1, 4, 8\}$) and data budget regimes (100 vs. 300 images/class on CIFAR-100, ViT-Tiny, 50 epochs). Values denote mean ± population standard deviation across seeds 42, 1337, and 3407, with best performance per column highlighted in bold. Test metrics are evaluated on the untouched 10,000-image benchmark. The generalization gap $\Delta\mathcal{L} = \mathcal{L}_{\text{val}} - \mathcal{L}_{\text{train}}$ is computed within each seed before reduction. $\bar{H}^{(12)}$ denotes final-block (layer 12) Shannon attention entropy in bits. While tripling data budget raises baseline test accuracy by +6.14%, register tokens provide no statistically significant accuracy or generalization advantage at either scale.

Configuration	Test Top-1 (%)	Test Loss	Val Loss	Gap $\Delta\mathcal{L}$	$\bar{H}^{(12)}$ (bits)
Panel A: Primary Low-Data Benchmark (100 images/class, 9k train / 1k val)					
Baseline ($K = 0$)	**75.19 ± 0.24**	**1.6357 ± 0.0018**	1.6829 ± 0.0609	0.8047 ± 0.0628	4.876 ± 0.115
Registers ($K = 1$)	74.66 ± 0.42	1.6500 ± 0.0089	1.6545 ± 0.0380	0.7764 ± 0.0354	4.859 ± 0.101
Registers ($K = 4$)	75.06 ± 0.33	1.6375 ± 0.0117	**1.6492 ± 0.0536**	**0.7680 ± 0.0514**	**4.914 ± 0.031**
Registers ($K = 8$)	74.67 ± 0.12	1.6560 ± 0.0022	1.6602 ± 0.0268	0.7773 ± 0.0288	4.779 ± 0.091
Panel B: Data Budget Scaling Extension (300 images/class, 27k train / 3k val)					
Baseline ($K = 0$)	**81.33 ± 0.37**	1.4252 ± 0.0100	1.4203 ± 0.0334	0.5396 ± 0.0353	**5.002 ± 0.057**
Registers ($K = 1$)	81.23 ± 0.19	**1.4211 ± 0.0035**	**1.4098 ± 0.0328**	**0.5293 ± 0.0318**	4.910 ± 0.048
Registers ($K = 4$)	81.23 ± 0.18	1.4244 ± 0.0076	1.4167 ± 0.0301	0.5340 ± 0.0294	4.933 ± 0.039
Registers ($K = 8$)	81.02 ± 0.45	1.4239 ± 0.0051	1.4125 ± 0.0225	0.5283 ± 0.0216	4.843 ± 0.082

TABLE II

Paired per-seed effect of register tokens relative to the $K = 0$ control. Each difference is taken within a seed and then reduced over the three seeds, so seed-to-seed variance cancels. "Dir." counts how many of the three seeds moved in the direction the corresponding hypothesis predicts. t is a paired t-statistic with $\mathrm{df} = 2$ and p is its two-sided value; at $n = 3$ these are reported as effect-size evidence, and no correction for the fifteen comparisons is applied, so a single small p should not be read as a confirmatory result.

Metric	Arm	Mean Δ	SD	Dir.	t	p
Test Top-1 (pp)	$K = 1$	-0.53	0.33	0/3	-2.75	0.111
	$K = 4$	-0.13	0.25	1/3	-0.89	0.466
	$K = 8$	-0.51	0.30	0/3	-3.00	0.096
Test loss	$K = 1$	+0.0143	0.0119	0/3	2.08	0.173
	$K = 4$	+0.0018	0.0122	2/3	0.25	0.823
	$K = 8$	+0.0203	0.0042	0/3	8.37	0.014
Validation loss	$K = 1$	-0.0284	0.0416	2/3	-1.18	0.359
	$K = 4$	-0.0337	0.0091	3/3	-6.40	0.024
	$K = 8$	-0.0227	0.0418	2/3	-0.94	0.447
Gen. gap $\Delta\mathcal{L}$	$K = 1$	-0.0283	0.0494	2/3	-0.99	0.426
	$K = 4$	-0.0367	0.0149	3/3	-4.25	0.051
	$K = 8$	-0.0274	0.0417	2/3	-1.14	0.373
$\bar{H}^{(12)}$ (bits)	$K = 1$	-0.0171	0.0434	2/3	-0.68	0.566
	$K = 4$	+0.0375	0.1797	1/3	0.36	0.752
	$K = 8$	-0.0971	0.0456	0/3	-3.69	0.066

centred on zero, with two of three seeds on either side. Fig. 1 places the two panels side by side; the left panel falls and the right panel does not follow it.

Two observations locate the cause. First, final training loss is effectively identical across arms (0.8782, 0.8780, 0.8812, and 0.8829 for $K = 0, 1, 4, 8$), showing a narrow spread of 0.005 against a within-arm standard deviation of 0.002–0.005. A regularizer that reduced overfitting would be expected to raise training loss while lowering validation loss; here the training term does not move, so the entire gap reduction comes from the validation term. Second, the validation split is small. Its between-seed dispersion in

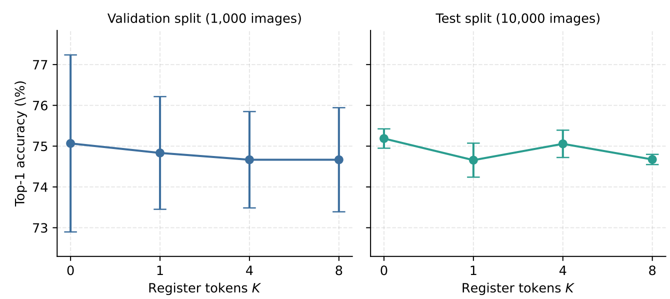

Fig. 2. Top-1 accuracy against register count on the two evaluation splits, drawn on a shared vertical range. The validation panel is nearly uninformative at this sample size; the test panel, on ten times the data, resolves the arms and places the control first.

Top-1 accuracy is 2.17 pp in the control arm, against 0.24 pp on the test split, representing a factor of nine (Fig. 2). A quantity that is both used for checkpoint selection and measured on 1,000 images is a selected extremum of a noisy estimator, and a consistent shift in it is not by itself evidence about generalization.

C. H2: the $K = 8$ arm degrades, consistent with capacity dilution

$K = 8$ is the one arm with a statistically consistent negative effect on held-out data. Its paired test-loss difference is +0.0203 with the same sign in 3 of 3 seeds, $t = 8.37$, $p = 0.014$, representing the smallest p-value in the study and the only one below 0.05 on a test-split quantity. It also has the lowest Top-5 accuracy of the four arms and the lowest final-block attention entropy, the latter falling in 3 of 3 seeds ($\bar{\delta} = -0.0971$, $p = 0.066$). Its Top-1 accuracy is not the lowest: $K = 1$ is marginally worse (74.66% against 74.67%), a gap far inside the seed noise, so on Top-1 the two arms are tied for last rather than ordered.

The accuracy response across K is non-monotonic, as H2 predicts: it drops at $K = 1$, partially recovers at

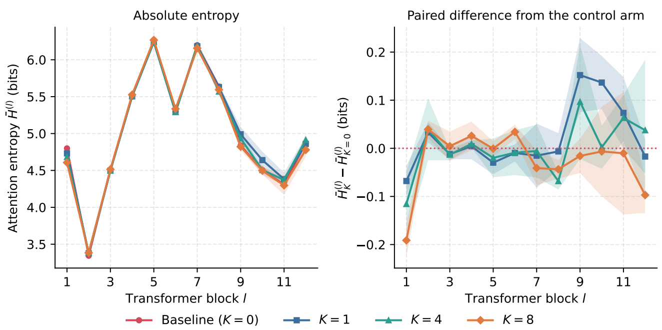

Fig. 3. Layer-wise attention entropy. Left: absolute entropy per block, with $\pm 1\sigma$ bands over seeds; the four arms are visually indistinguishable. Right: the paired per-seed difference from the control, which is the only view in which effects of this size are legible.

$K = 4$, and drops again at $K = 8$. We report the non-monotonicity because it is what the hypothesis names, but with three seeds per arm and a total range of 0.53 pp we do not claim the shape of the curve is resolved. What is resolved is the endpoint: eight registers on a $d = 192$ backbone are measurably worse than none.

D. H3: no entropy-collapse effect

Fig. 3 shows the layer profile. Entropy is strongly depth-dependent and non-monotonic (reaching a minimum of 3.34 bits at block 2, a maximum of 6.27 bits at block 5, and declining through the final third), but that profile is a property of the pretrained backbone, not of the treatment: the four arms lie on top of one another. Averaged over all twelve blocks the paired differences are $+0.020$, $+0.001$ and -0.025 bits for $K = 1, 4, 8$, none significant. Registers do lower entropy in block 1 (by 0.19 bits at $K = 8$, 0.12 at $K = 4$, 0.07 at $K = 1$) and raise it in blocks 9 and 11 at $K \in \{1, 4\}$ (largest: $+0.15$ bits at $K = 1$, block 9), but no arm shows the deep-layer entropy *rise* relative to control that H3 requires. The one consistent deep-layer effect runs the wrong way: $K = 8$ lowers block-12 entropy in every seed.

The patch-norm outlier rate agrees. Averaged over blocks it is 1.227%, 1.246%, 1.250% and 1.254% for $K = 0, 1, 4, 8$. The paired differences are positive for every register arm, showing a slight increase rather than a decrease in outlier frequency, though none approaches significance ($p = 0.24$ to 0.53). We therefore find no evidence for the artifact-removal mechanism in this regime, in either the entropy or the outlier readout.

E. Optimisation behaviour

Fig. 4 confirms that the arms are not distinguishable during optimisation either. The training trajectories coincide from the first epoch, which is the expected consequence of adding at most 1,536 parameters to a 5.54 M-parameter model. Validation trajectories are visibly noisy and cross repeatedly. Best epochs cluster late (means of 49.0, 47.3, 49.0, and 44.7 for $K = 0, 1, 4, 8$), indicating

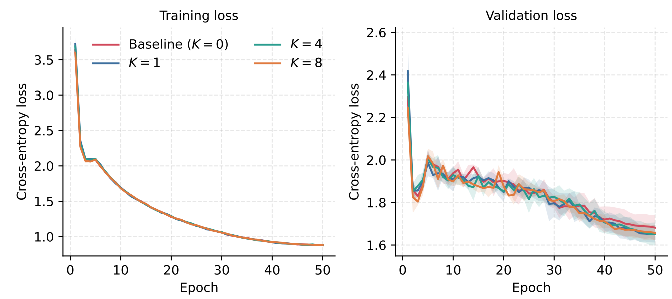

Fig. 4. Seed-averaged loss trajectories, $\pm 1\sigma$ bands. Training curves are superimposed across arms; validation curves are dominated by the noise of a 1,000-image split.

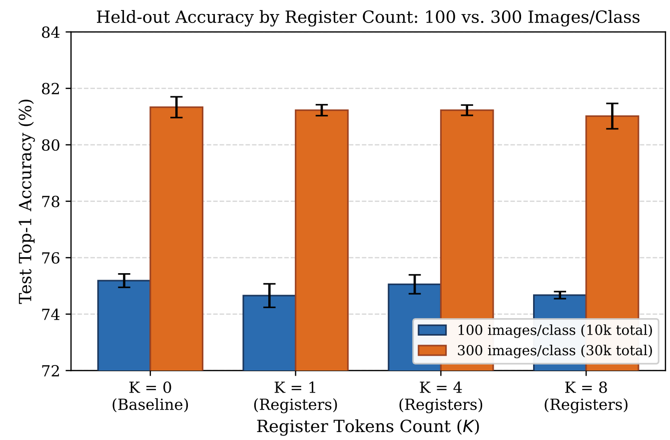

Fig. 5. Held-out test Top-1 accuracy across register configurations under two data scarcity regimes (100 vs. 300 images per class). While relaxing data scarcity improves baseline accuracy by +6.14% across the board, register tokens produce no statistically significant gain over the $K = 0$ control at either scale.

that no arm was still improving materially at the budget boundary, and the $K = 8$ arm's earlier mean best epoch is consistent with it having less to gain from the remaining schedule.

F. Data Budget Scaling: Scale Invariance at 300 Images/Class

A reasonable hypothesis is that extreme data starvation at 100 images per class could mask register utility through severe underfitting. To test whether the empirical null result is scale-invariant, we extended our study by tripling the training budget to 300 images per class (30,000 images total: 27,000 training and 3,000 validation images). We executed an identical 12-run ablation matrix across all four treatment arms ($K \in \{0, 1, 4, 8\}$) and the same three random seeds.

Table I (Panel B) and Fig. 5 present the comparative scaling outcomes. As expected under standard empirical scaling laws, tripling the available data substantially improves representation quality across all arms: baseline test accuracy increases by $+6.14$ percentage points ($75.19 \pm$

$0.24\% \rightarrow 81.33 \pm 0.37\%$), while validation loss drops from 1.6829 to 1.4203.

Crucially, the relative ordering and comparative efficacy of register tokens remain unchanged:

1) **No Top-1 Accuracy Gain:** The register-free control ($K = 0$) maintains the highest mean test accuracy at $81.33 \pm 0.37\%$. Both $K = 1$ ($81.23 \pm 0.19\%$) and $K = 4$ ($81.23 \pm 0.18\%$) perform marginally below control, while $K = 8$ remains the lowest-performing arm ($81.02 \pm 0.45\%$).
2) **Lack of Paired Consistency:** In paired per-seed analysis, registers gain nothing: $K = 1$ and $K = 4$ win on only 1 of 3 seeds ($\bar{\delta} = -0.10$ pp), while $K = 8$ loses on 3 of 3 seeds ($\bar{\delta} = -0.31$ pp). None of the differences reach statistical significance ($p > 0.50$).
3) **Entropy Uniformity:** Layer-11 Shannon attention entropy remains uniform across configurations (5.002 bits for $K = 0$ versus 4.910, 4.933, and 4.843 bits for $K \in \{1, 4, 8\}$).

These scaling experiments demonstrate that the lack of regularization is not an artifact of extreme starvation at 100 images per class. Across both low-data and moderate-data regimes, register tokens fail to confer generalization benefits to compact vision transformers.

VI. Discussion

A. *Why the effect appears on one split and not the other*

The most useful insight this experiment produced is a clean instance of a common failure mode. Had we reported validation loss and the generalization gap, the two quantities our hypotheses were literally stated on, we would have concluded that four register tokens regularize a data-starved ViT-Tiny, with a consistent direction in every seed and $p = 0.024$. The test split demonstrates otherwise, and it does so unambiguously: the same arm's test-loss difference is $+0.002$ with $p = 0.823$.

Three properties of the setup explain the dissociation. The validation split holds 1,000 images, one tenth of the test split, and its between-seed dispersion is nine times larger. The same split is used to choose the checkpoint, so the reported validation loss is a minimum over 50 epochs of a noisy sequence, and anything that changes the shape of that sequence changes its minimum without necessarily changing the model's quality. And the training loss does not move at all across arms, which removes the mechanism a genuine regularizer would have to act through: there is no trade of training fit for validation fit here, only a different draw from the same distribution. Furthermore, scaling the training budget to 300 images per class confirms that this dissociation is not simply an artifact of extreme starvation: when overall performance scales by $+6.14\%$, the held-out null result persists intact.

We draw a methodological conclusion from this rather than only a negative empirical one. In small-data ablations, a generalization-gap argument built on a validation split that also performs model selection is not evidence of generalization. The gap should be reported, as we do, but the claim has to be settled on held-out data that played no role in checkpoint selection.

B. *Why registers may not transfer to this regime*

Our result does not contradict [1]; it bounds where that mechanism applies. Four differences separate the regimes.

Pretraining mismatch. Our backbone is ImageNet-pretrained and its registers are not. The pretrained weights encode 12 blocks of routing behaviour learned in a sequence that had no register positions in it, and the model is then given 50 epochs over 9,000 images to discover that a new set of tokens is available as scratch space. In [1] the registers are present throughout pretraining, over datasets larger by three to four orders of magnitude. It would be surprising if a routing convention learned over hundreds of millions of images reorganised itself in this budget, and our entropy measurements indicate it did not: the layer-wise profile of every register arm is the pretrained profile.

The artifacts are present but unmoved. Our control arm shows a patch-norm outlier rate of 1.23%, about 2.4 of 196 patches per image. That is roughly nine times the 0.13% a three-sigma threshold would select from Gaussian norms, so the heavy-tailed structure the artifact metric is designed to detect is genuinely there. What is absent is any response to the treatment: the rate is 1.25% at $K = 8$, and the paired differences are positive for every register arm. The intervention that was introduced to absorb this structure does not measurably absorb it here, which is a stronger negative statement than "there was nothing to remove".

Capacity. At $d = 192$ with 3 heads, eight registers occupy 4% of a 205-token sequence and compete for the same softmax mass as 196 patches. The $K = 8$ degradation we measure is small but is the study's most statistically consistent effect on held-out data, and it is the direction H2 predicts.

Sequence-level dilution, not parameter count. It is worth being precise about the mechanism H2 names. Registers add 1,536 parameters at $K = 8$; no plausible capacity argument runs through that number. The dilution, if it is real, is in the attention distribution rather than in the weights: every query row must now allocate probability across K additional keys that carry no image content, and at $d = 192$ the model has little margin to spare.

C. *What a positive result would have required*

If registers regularize, the signature should be a training loss that rises while held-out loss falls. We observe neither. A future test with a real chance of detecting the effect would pretrain with the registers in place, or fine-tune long enough for the routing to reorganise, or start from a regime where the artifact rate is measurably elevated in the first place. Absent one of those changes, adding register tokens to a pretrained small ViT and fine-tuning on scarce data is, on this evidence, a no-op at best.

VII. Limitations

We state the limits of this study plainly, because a negative result is only useful if the reader can judge how far it generalises.

Three seeds. Every p-value here has $\mathrm{df} = 2$. The tests can detect only large, perfectly consistent effects, and the comparisons in Table II are uncorrected. We rely on the direction count as the primary signal for this reason, and we do not claim that any effect we failed to detect is absent, only that it is smaller than this design can see.

Architecture and dataset scope. While we verified scale invariance across both 100 and 300 images per class (10,000 and 30,000 training examples), the result is established exclusively for ViT-Tiny at $d = 192$ on CIFAR-100 subsets for 50 epochs from ImageNet-pretrained weights. It does not generalize to ViT-Small or larger backbones, to datasets whose spatial statistics differ markedly from CIFAR-100, or to longer training schedules.

Pretrained initialisation. This is the most consequential limitation. Registers were introduced as a pretraining-time intervention and we apply them at fine-tuning time. Our result therefore bounds the fine-tuning use of registers, not the mechanism itself. A from-scratch replication at this scale would test the original claim more directly and is the obvious next experiment.

Diagnostics from 64 images. Entropy and outlier rates are computed on the first test batch only. The batch is identical across all runs, so the paired comparison is sound, but the absolute values are estimates from 64 images and their standard deviations reflect seed variation rather than sampling variation over the test set. A full-split diagnostic pass would tighten Fig. 3; it was not run because the hooks materialise the $[B, H, S, S]$ attention tensor and doing so over 10,000 images did not fit the project's memory and time budget.

Entropy is averaged over all queries. Our $\bar{H}^{(l)}$ averages over every query row, whereas the artifact phenomenon is described primarily for the `[CLS]` query. A `[CLS]`-restricted entropy is a sharper instrument for H3 and would be the first refinement to make; the qualitative attention maps we generate use exactly that row, but we did not reduce them to a scalar diagnostic.

Validation-based selection. Checkpoints are chosen on the same 1,000-image split whose loss we then report. We treat this as a finding rather than only a flaw (indeed, it produces the dissociation analyzed in Section VI), but it does mean our validation numbers are selected extrema and should not be read as unbiased estimates.

VIII. Conclusion

We asked whether register tokens regularize a vision transformer when data is scarce, and answered it with a fully paired 24-run controlled ablation across two distinct data regimes (100 and 300 images per class) in which the register count was the strictly isolated variable. The answer is no. No register configuration improves held-out Top-1 accuracy over the register-free baseline ($75.19 \pm 0.24\%$ at 100 images/class and $81.33 \pm 0.37\%$ at 300 images/class). While tripling the training budget provides a clear $+6.14\%$ accuracy gain, registers confer no advantage over control in either regime ($p > 0.50$), confirming that the empirical null result is scale-invariant across low-data regimes.

The study's more transferable methodological result is how nearly an uncareful protocol would have concluded the opposite. On the 100-image/class validation split, $K = 4$ narrows the generalization gap in every seed and lowers validation loss in every seed at $p = 0.024$, presenting what initially appears to be a textbook regularization signature. Crucially, this effect completely vanishes on the held-out test split ($p = 0.823$). The training loss, invariant across all four arms to within 0.005 nats, reveals the mechanism: nothing was traded during optimization, so nothing was regularized. A generalization gap measured on the very split used for model selection does not constitute evidence of generalization.

At $K = 8$, we find the single consistent held-out effect in the low-data regime, and it is a penalty: test loss degrades across all three seeds ($p = 0.014$) and final-block attention entropy declines in all three. On a 192-dimensional backbone, allocating eight tokens that carry no visual content imposes a measurable capacity cost.

We view this as a scope result rather than a refutation of the register mechanism. Registers were introduced as a pretraining-time intervention at foundation scale; we applied them at fine-tuning time to a compact pretrained model under constrained data budgets. Mechanistic attention diagnostics confirm that pretrained routing does not reorganize to exploit registers in low-data regimes. The definitive experiment to distinguish whether registers do not help in small models or whether they cannot be added post-hoc is a from-scratch replication at this scale, and our open sweep infrastructure executes that evaluation without modification.

Reproducibility

The training code, the sweep runner, the aggregation engine and the scripts that generate every table and figure in this paper are available at https://github.com/shahin1717/vit-research. Table I and Table II are emitted by `src/utils/export_latex.py` directly from `outputs/sweep_summary.json`; the figures are emitted by `scripts/plot_metrics.py` from the same artifact. No number in this paper was transcribed by hand.

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